

# Olympiad Geometry – VII

## Inversion

The most exotic geometric transformation we shall cover is inversion.

Unlike the transformations we have seen so far, when applying inversion, figures may substantially change their shape. Yet, as we will see, inversion is an ultimately powerful tool in solving geometric problems.

Properties of inversion can be stated more efficiently if we introduce a point at infinity. We shall denote it as  $\infty$  and we establish that it lies on each line. This extended plane is called the inversion plane.

## Definition

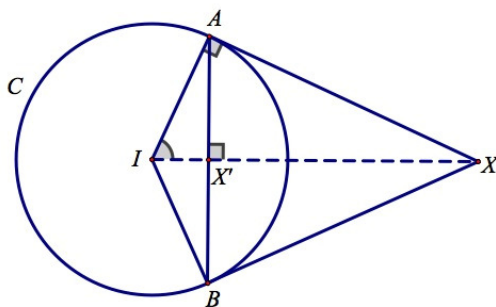
Take your favourite circle  $C$  with centre  $I$  and radius  $r > 0$ .

We define the image  $X'$  of point  $X$  under inversion about  $C$  as follows:

- (1) If  $X = I$ , then  $X' = \infty$ .
- (2) If  $X = \infty$ , then  $X' = I$ .
- (3) Otherwise,  $X'$  is such point on ray  $IX$  such that  $IX \cdot IX' = r^2$ .

Observe that points inside  $C$  (with  $IX < r$ ) are mapped to the outside ( $IX' > r$ ) and vice versa, while  $C$  is left intact. Further, if we perform inversion about the same circle twice, we obtain identity mapping (nothing happens). In other words,  $X'$  is the image of  $X$  if and only if  $X$  is the image of  $X'$ .

Let  $X$  be a point outside the circle  $C$  centred at  $I$ . Let tangents from  $X$  touch  $C$  at points  $A, B$ . Finally, denote by  $X'$  the midpoint of  $AB$ . Then  $X'$  is the image of  $X$  under inversion about  $C$ .

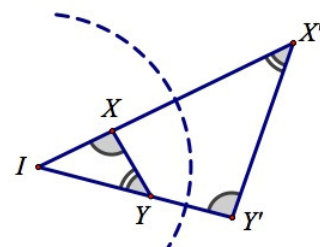


## Theorem

Let  $I, X, Y$  be pairwise distinct non-collinear points. Denote by  $X'$  and  $Y'$  the images of  $X$  and  $Y$  under inversion about a circle with centre  $I$  and radius  $r > 0$ .

Then  $\triangle XIY \sim \triangle Y'IX'$  with ratio of similitude  $\frac{XY'}{XY} = \frac{r^2}{IX \cdot IY}$ . In particular,

- (a)  $\angle XYI = \angle IX'Y'$ .
- (b)  $XY' = XY \cdot \frac{r^2}{IX \cdot IY}$ .
- (c)  $XY = XY' \cdot \frac{r^2}{IX' \cdot IY'}$ .



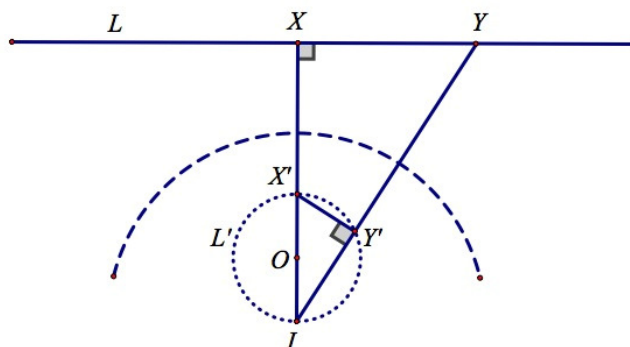
As we will see, the radius of inversion may often be chosen arbitrarily. In such case, we shall use the notion of inverting about a point. The radius will be considered to be equal to 1.

Now let's see what happens to lines and circles after inversion. The answer is surprisingly convenient!

**Image of lines under inversion**

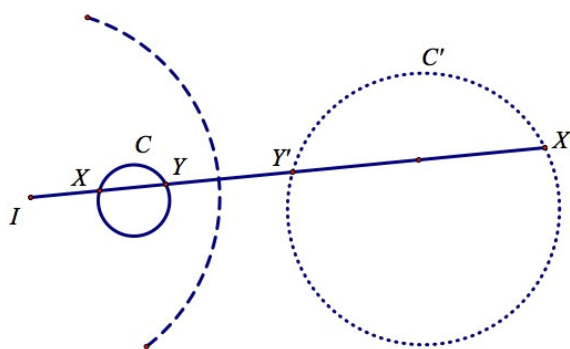
Denote by  $L'$  the image of line  $L$  under inversion about  $I$ .

- (a) If  $I \in L$ , then  $L' = L$ .
- (b) If  $I \notin L$ , then  $L'$  is a circle with centre  $O$  passing through  $I$  such that  $OI \perp L$ .

**Image of circles under inversion**

Denote by  $C'$  the image of circle  $C$  with centre  $O$  under inversion about  $I$ .

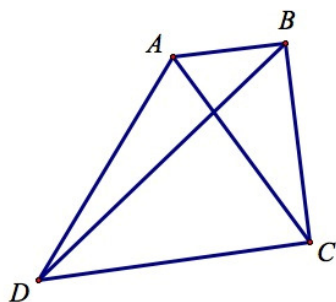
- (a) If  $I \in C$ , then  $C'$  is a line perpendicular to  $OI$ .
- (b) If  $I \notin C$ , then  $C'$  is a circle. Moreover, centres of  $C$  and  $C'$  are collinear with  $I$ .



The idea of using inversion is that we invert both the figure and the desired conclusion to obtain an equivalent problem. Very often (but not always!) this equivalent problem is far easier to solve. As we will see that inverting about a point with many circles passing through it usually leads to much simpler figure.

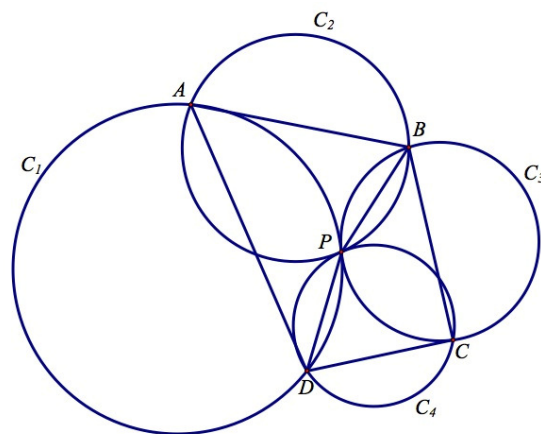
**Example 1 (Ptolemy's Inequality)**

The mutual distances of four distinct points  $A, B, C, D$  satisfy  $AB \times CD + BC \times AD \geq AC \times BD$  with the equals sign only when  $A, B, C, D$  lie on a circle (or on a line) in such an order that either of the arcs  $AC$  (or the line segment  $AC$ ) contains one but not both of the remaining points  $B$  and  $D$ .

**Example 2 (IMO 2003 shortlist)**

Let  $C_1, C_2, C_3, C_4$  be distinct circles such that  $C_1, C_3$  are externally tangent at  $P$ , and  $C_2, C_4$  are externally tangent at the same point  $P$ . Suppose that  $C_1$  and  $C_2, C_2$  and  $C_3, C_3$  and  $C_4, C_4$  and  $C_1$  meet at  $A, B, C, D$  respectively, and that all these points are different from  $P$ . Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}$$

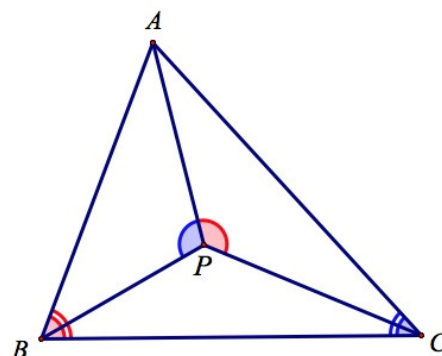


**Example 3 (IMO 1996)**

Let  $P$  be a point inside a triangle  $ABC$  such that  $\angle APB - \angle ACB = \angle APC - \angle ABC$ .

Let  $D, E$  be the incentres of triangles  $APB, APC$ , respectively.

Show that the lines  $AP, BD, CE$  meet a point.

**Example 4 (APMO 1994)**

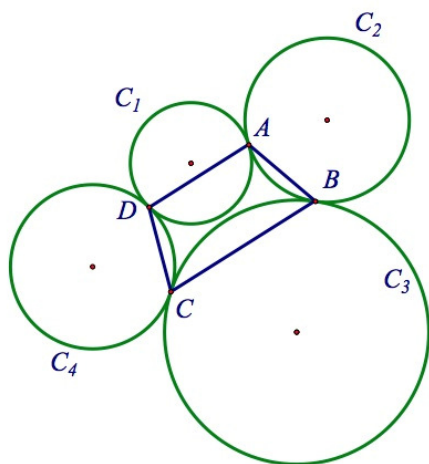
Is there an infinite set of points of the plane such that no three elements of it are collinear, and the distance between any two of them is rational?



## Problems

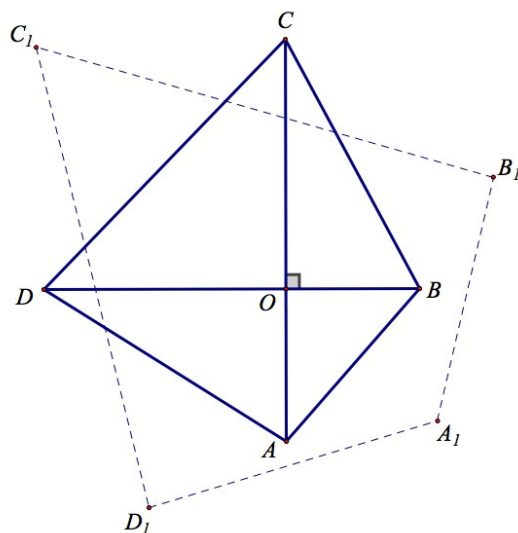
### Problem 1

Circles  $C_1$ ,  $C_2$ ,  $C_3$  and  $C_4$  are positioned in a such a way that  $C_1$  is tangent to  $C_2$  at point  $A$ ,  $C_2$  is tangent to  $C_3$  at point  $B$ ,  $C_3$  is tangent to  $C_4$  at point  $C$ , and  $C_4$  is tangent to  $C_1$  at point  $D$ . Show that  $A$ ,  $B$ ,  $C$  and  $D$  are either collinear or concyclic.



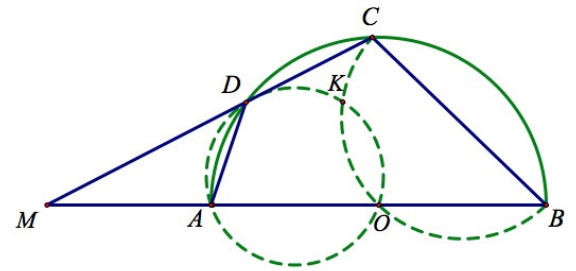
### Problem 2 (1993 USAMO)

Let  $ABCD$  be a convex quadrilateral whose diagonals intersect at right angles at  $O$ . Prove that the reflections of  $O$  across  $AB$ ,  $BC$ ,  $CD$ ,  $DA$  are concyclic.



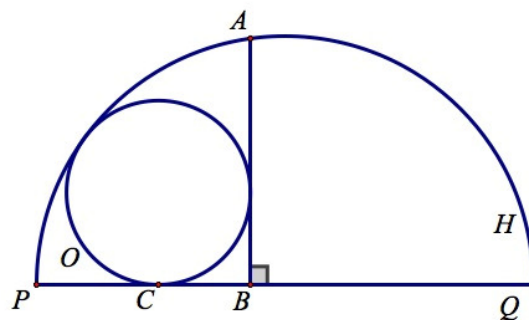
**Problem 3 (1995 Russian Math Olympiad)**

Let  $ABCD$  be a quadrilateral inscribed in a semicircle with diameter  $AB$  and centre  $O$ . Lines  $CD$  and  $AB$  intersect at  $M$ . Let  $K$  be the second point of intersection of the circumcircles of triangles  $AOD$  and  $COB$ . Prove that  $\angle MKO = 90^\circ$ .



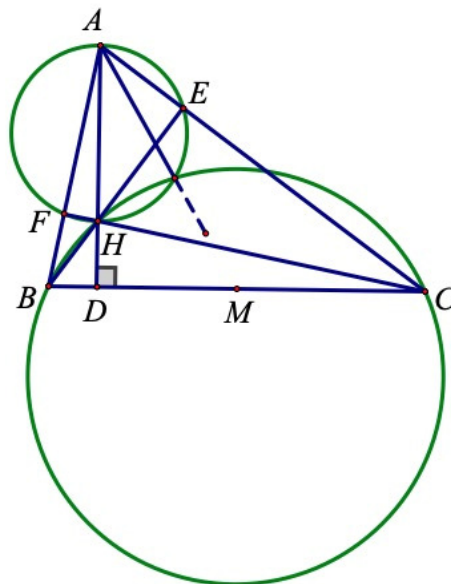
**Problem 4 (1995 Israeli Math Olympiad)**

Let  $PQ$  be the diameter of semicircle  $H$ . Circle  $O$  is internally tangent to  $H$  and tangent to  $PQ$  at  $C$ . Let  $A$  be a point on  $H$  and  $B$  a point on  $PQ$  such that  $AB \perp PQ$  and is tangent to  $O$ . Prove that  $AC$  bisects  $\angle PAB$ .



**Problem 5**

In  $\triangle ABC$  with orthcentre  $H$ , the circle with diameter  $AH$  and the circumcircle  $\triangle BHC$  intersect again on the  $A$ -median.



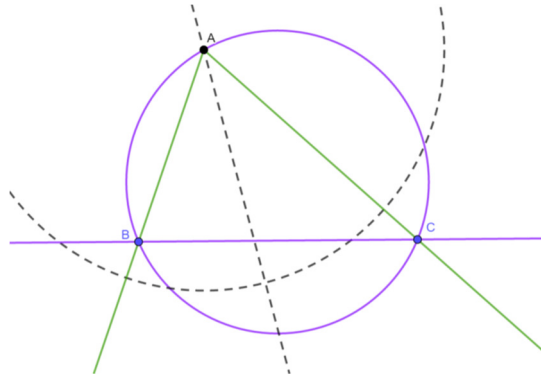


$\sqrt{bc}$  -inversion

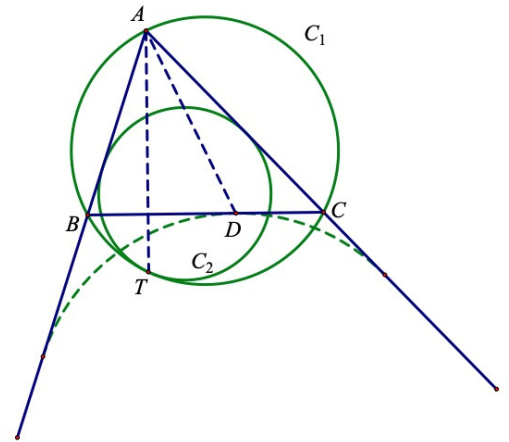
Given a  $\triangle ABC$ , we consider the transformation which first reflects point  $X$  over the  $A$ -angle bisector, then inverts about  $A$  with radius  $\sqrt{AB \cdot AC}$  into  $X'$  the image of  $X$  in  $\sqrt{bc}$ -inversion. Complicated at the first glance, but such a trick is “tailor-made” to the triangle such that

1.  $B$  maps to  $C$  and  $C$  maps to  $B$ ; and
2. the circumcircle of  $\triangle ABC$  maps to line  $BC$ , line  $BC$  maps to the circumcircle of  $\triangle ABC$ .

Note that lines  $AX$  and  $AX'$  are isogonal, so it is comfy to use such a trick in isogonal lines.

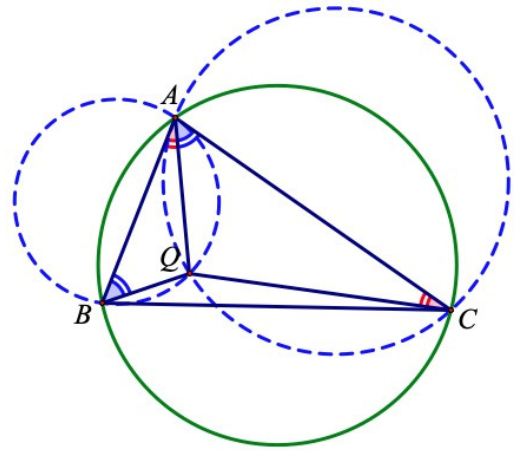
**Example 5 (Mixtilinear Incircle)**

Let  $C_1$  be the circumcircle of  $\triangle ABC$ . Circle  $C_2$  is tangent to  $AB$  and  $AC$ , and it touches  $C_1$  internally at  $T$ . Let  $D$  be the point of tangency of  $BC$  and the  $A$ -excircle. Show that  $\angle BAT = \angle DAC$ .



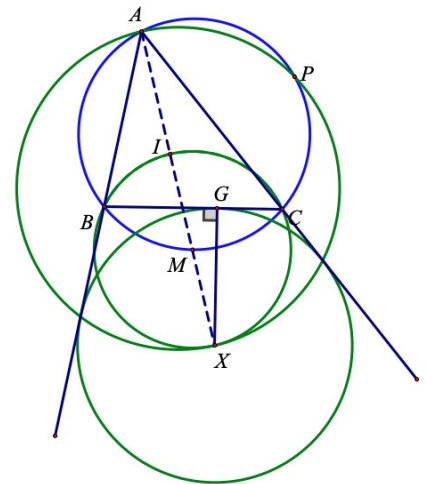
**Example 6 (Dumpty-point)**

In  $\triangle ABC$ , let  $S$  be a point on the circumcircle of  $\triangle ABC$  such that  $AS$  is the  $A$ -symmedian. Prove that the midpoint of  $AS$  is exactly the  $A$ -dumpty point, i.e. the point  $Q$  satisfying  $\angle BAQ = \angle ACQ$  and  $\angle CAQ = \angle ABQ$ .



**Example 7**

Let  $ABC$  be a triangle where  $X$  is the  $A$ -excentre and  $I$  is the incentre. Let  $M$  be the circumcentre of  $\triangle BIC$ , and let  $G$  be the point on  $BC$  such that  $XG \perp BC$ . Construct a circle with diameter  $AX$ . If the circle with diameter  $AX$  and the circumcircle of  $\triangle ABC$  intersect at a second point  $P$ , prove that  $M$ ,  $G$  and  $P$  are collinear.



**Harder Problem (IMO 2015)**

Let  $ABC$  be an acute triangle with  $AB > AC$ . Let  $\Gamma$  be its circumcircle,  $H$  its orthocentre, and  $F$  the foot of the altitude from  $A$ . Let  $M$  be the midpoint of  $BC$ . Let  $Q$  be the point on  $\Gamma$  such that  $\angle HQA = 90^\circ$  and let  $K$  be the point on  $\Gamma$  such that  $\angle HKQ = 90^\circ$ . Assume that the points  $A, B, C, K$  and  $Q$  are all different and lie on  $\Gamma$  in this order. Prove that the circumcircles of triangles  $KQH$  and  $FKM$  are tangent to each other.

